



CS39006: NETWORKS LABORATORY

ASSIGNMENT 1: NETWORKING TOOLS AND WIRESHARK

NAME: Tuhin Mondal
ROLL NO: 22CS10087

PART – 1: NETWORKING TOOLS

1) Find the IP address of your machine, subnet mask, and network ID of your subnet.

```
user@swlab-PC:~$ ifconfig
eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.5.16.134 netmask 255.255.255.0 broadcast 10.5.16.255
    inet6 fe80::dbe0:9767:583:ff32 prefixlen 64 scopeid 0x20<link>
    ether 84:a9:3e:69:f0:2c txqueuelen 1000 (Ethernet)
    RX packets 83097 bytes 52246779 (52.2 MB)
    RX errors 0 dropped 70 overruns 0 frame 0
    TX packets 14258 bytes 1881581 (1.8 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xf1000000-f1020000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 3142 bytes 753172 (753.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3142 bytes 753172 (753.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp0s20f3: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 40:74:e0:b6:23:26 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

user@swlab-PC:~$
```

IP Address: 10.5.16.134
Subnet Mask: 255.255.255.0
Network ID: 10.5.16.0

Identified the IP address and subnet mask from the output of ifconfig command. We can calculate the Network ID by performing a bitwise AND operation between the IP address and subnet mask:

```
- IP Address (binary): 00001010.00000101.00010000.10000110
- Subnet Mask (binary): 11111111.11111111.11111111.00000000
=====
- Network ID (binary): 00001010.00000101.00010000.00000000
```

2) Find the IP address associated with www.google.com and www.facebook.com using nslookup. Change the DNS server address in the nslookup command to the following four IP addresses: 172.16.1.164, 172.16.1.180, 172.16.1.165, and 172.16.1.166, and see whether the IP address of the above domain name(www.google.com) changes. If you see a change in the IP address of www.google.com, can you think of the reason behind the same?

DNS Server Responses:

The IP addresses for www.google.com differ based on the DNS server queried:

172.16.1.164: 142.250.192.68 (IPv4), 2404:6800:4009:825::2004 (IPv6)
172.16.1.180: 142.250.206.164 (IPv4), 2404:6800:4002:81a::2004 (IPv6)
172.16.1.165: 142.250.192.100 (IPv4), 2404:6800:4009:82a::2004 (IPv6)
172.16.1.166: 142.250.77.36 (IPv4), 2404:6800:4009:81c::2004 (IPv6)

The IP addresses for www.facebook.com differ based on the DNS server queried:

172.16.1.166: 31.13.79.35 (IPv4), 2a03:2880:f12f:183:face:b00c:0:25de (IPv6)
172.16.1.164: 31.13.79.35 (IPv4), 2a03:2880:f12f:183:face:b00c:0:25de (IPv6)
172.16.1.165: 31.13.79.35 (IPv4), 2a03:2880:f12f:183:face:b00c:0:25de (IPv6)
172.16.1.180: 157.240.16.35 (IPv4), 2a03:2880:f10c:181:face:b00c:0:25de (IPv6)

Summary:

DNS Server	Domain	IPv4 Address	IPv6 Address
172.16.1.164	www.google.com	142.250.192.68	2404:6800:4009:825::2004
172.16.1.180	www.google.com	142.250.206.164	2404:6800:4002:81a::2004
172.16.1.165	www.google.com	142.250.192.100	2404:6800:4009:82a::2004
172.16.1.166	www.google.com	142.250.77.36	2404:6800:4009:81c::2004
172.16.1.166	www.facebook.com	31.13.79.35	2a03:2880:f12f:183:face:b00c:0:25de
172.16.1.164	www.facebook.com	31.13.79.35	2a03:2880:f12f:183:face:b00c:0:25de
172.16.1.165	www.facebook.com	31.13.79.35	2a03:2880:f12f:183:face:b00c:0:25de
172.16.1.180	www.facebook.com	157.240.16.35	2a03:2880:f10c:181:face:b00c:0:25de

- **tuhin@tuhin-hp15s-fy5xxx:~\$ nslookup www.google.com 172.16.1.164**

```
Server:      172.16.1.164
Address:     172.16.1.164#53
```

Non-authoritative answer:

```
Name:   www.google.com
Address: 142.250.192.4
Name:   www.google.com
Address: 2404:6800:4009:82c::2004
```

- **tuhin@tuhin-hp15s-fy5xxx:~\$ nslookup www.google.com 172.16.1.180**

```
Server:      172.16.1.180
Address:     172.16.1.180#53
```

Non-authoritative answer:

```
Name:   www.google.com
Address: 142.250.192.196
Name:   www.google.com
Address: 2404:6800:4002:81a::2004
```

- **tuhin@tuhin-hp15s-fy5xxx:~\$ nslookup www.google.com 172.16.1.165**

```
Server:      172.16.1.165
Address:     172.16.1.165#53
```

Non-authoritative answer:

```
Name:   www.google.com
Address: 142.250.192.100
Name:   www.google.com
Address: 2404:6800:4009:82a::2004
```

- **tuhin@tuhin-hp15s-fy5xxx:~\$ nslookup www.google.com 172.16.1.166**

```
Server:      172.16.1.166
Address:     172.16.1.166#53
```

Non-authoritative answer:

```
Name:   www.google.com
Address: 142.250.77.36
Name:   www.google.com
Address: 2404:6800:4009:81c::2004
```

```

● tuhin@tuhih-hp15s-fy5xxx:~$ nslookup www.facebook.com 172.16.1.166
Server:      172.16.1.166
Address:     172.16.1.166#53

Non-authoritative answer:
www.facebook.com      canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

● tuhin@tuhih-hp15s-fy5xxx:~$ nslookup www.facebook.com 172.16.1.164
Server:      172.16.1.164
Address:     172.16.1.164#53

Non-authoritative answer:
www.facebook.com      canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

● tuhin@tuhih-hp15s-fy5xxx:~$ nslookup www.facebook.com 172.16.1.165
Server:      172.16.1.165
Address:     172.16.1.165#53

Non-authoritative answer:
www.facebook.com      canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

● tuhin@tuhih-hp15s-fy5xxx:~$ nslookup www.facebook.com 172.16.1.180
Server:      172.16.1.180
Address:     172.16.1.180#53

Non-authoritative answer:
www.facebook.com      canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 157.240.16.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f10c:181:face:b00c:0:25de

```

Reason for Variations:

When we look up a website like www.google.com or www.facebook.com, the IP address you get can vary depending on the DNS server you use. This happens for a couple of important reasons:

(1) Load Balancing: Big companies like Google and Facebook don't rely on a single server to handle all their traffic. Instead, they use many servers spread across the globe to ensure their websites stay fast and available no matter where you are. When we type a website into your browser, the DNS server finds the best IP address for us to connect to. This choice depends on several factors, such as how busy a server is or how close it is to us. So, if we ask different DNS servers, you might get different answers because each server routes traffic to the IPs it thinks are most optimal at that moment. It's like choosing the fastest or least crowded road during rush hour.

(2) Regional Optimization: DNS servers often return results based on your geographic location. For example: If we're in Asia, your DNS query might be directed to servers closer to us in that region. If we're in Europe, we'll likely get IPs for servers based there. Additionally, DNS servers can have cached data, meaning they remember previous queries. Since the cached responses might come from different regions or times, this can also cause variation in the IPs they return.

All this ensures:

- **Faster loading:** By connecting you to the nearest or least-busy server.
- **High availability:** If one server is down or overloaded, another can take over.
- **Efficient traffic management:** Balancing millions (or even billions) of users across many servers.

3) Ping the IP address of one of your friend's machine IP within the software lab. Send the ping packets with different packet sizes (64, 128, 512 bytes) and timeout (100) for reporting packet loss percentage, min, avg, max, and stddev of round-trip time.

```
user@swlab-PC: ~$ ping 10.5.16.135 -s 64 -w 100
```

```
PING 10.5.16.135 (10.5.16.135) 64(92) bytes of data.
```

```
...
```

```
--- 10.5.16.135 ping statistics ---
```

```
98 packets transmitted, 98 received, 0% packet loss, time 99314ms
```

```
rtt min/avg/max/mdev = 0.332/0.647/0.768/0.099 ms
```

```
user@swlab-PC:~$ ping 10.5.16.135 -s 128 -w 100
```

```
PING 10.5.16.135 (10.5.16.135) 128(156) bytes of data.
```

```
...
```

```
--- 10.5.16.135 ping statistics ---
```

```
98 packets transmitted, 98 received, 0% packet loss, time 99265ms
```

```
rtt min/avg/max/mdev = 0.319/0.658/0.879/0.099 ms
```

```
user@swlab-PC:~$ ping 10.5.16.135 -s 512 -w 100
```

```
PING 10.5.16.135 (10.5.16.135) 512(540) bytes of data.
```

```
...
```

```
--- 10.5.16.135 ping statistics ---
```

```
98 packets transmitted, 98 received, 0% packet loss, time 99289ms
```

```
rtt min/avg/max/mdev = 0.284/0.618/0.851/0.133 ms
```

Packet Size (bytes)	Packets Transmitted	Packets Received	Packet Loss (%)	Total Time (ms)	RTT Min (ms)	RTT Avg (ms)	RTT Max (ms)	RTT SDev (ms)
64	98	98	0	99314	0.332	0.647	0.768	0.099
128	98	98	0	99265	0.319	0.658	0.879	0.099
512	98	98	0	99289	0.284	0.618	0.851	0.133

```
user@swlab-PC:~$ ping 10.5.16.135 -s 64 -c 1
```

```
PING 10.5.16.135 (10.5.16.135) 64(92) bytes of data.
```

```
72 bytes from 10.5.16.135: icmp_seq=1 ttl=64 time=0.530 ms
```

```
--- 10.5.16.135 ping statistics ---
```

```
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

```
rtt min/avg/max/mdev = 0.530/0.530/0.530/0.000 ms
```

```
user@swlab-PC:~$ ping 10.5.16.135 -s 128 -c 1
```

```
PING 10.5.16.135 (10.5.16.135) 128(156) bytes of data.
```

```
136 bytes from 10.5.16.135: icmp_seq=1 ttl=64 time=0.599 ms
```

```
--- 10.5.16.135 ping statistics ---
```

```
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

```
rtt min/avg/max/mdev = 0.599/0.599/0.599/0.000 ms
```

```
user@swlab-PC:~$ ping 10.5.16.135 -s 512 -c 1
```

```
PING 10.5.16.135 (10.5.16.135) 512(540) bytes of data.
```

```
520 bytes from 10.5.16.135: icmp_seq=1 ttl=64 time=0.661 ms
```

```
--- 10.5.16.135 ping statistics ---
```

```
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

```
rtt min/avg/max/mdev = 0.661/0.661/0.661/0.000 ms
```

04) Run traceroute for www.google.com and print the summary. Count the number of hosts involved in the path from source to destination. Why do you see some “* * *” in the intermediate hops?

```
user@swlab-PC:~$ traceroute www.google.com
traceroute to www.google.com (142.250.194.36), 30 hops max, 60 byte packets
 1 _gateway (10.5.16.2) 0.365 ms 0.305 ms 0.379 ms
 2 10.120.2.33 (10.120.2.33) 0.335 ms 0.291 ms 0.247 ms
 3 10.255.1.3 (10.255.1.3) 3.439 ms 3.866 ms 5.132 ms
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 72.14.204.62 (72.14.204.62) 46.238 ms 142.250.172.80 (142.250.172.80) 57.193 ms 50.202 ms
 9 * * *
10 * 216.239.50.166 (216.239.50.166) 47.767 ms 172.253.50.146 (172.253.50.146) 55.445 ms
11 192.178.110.110 (192.178.110.110) 55.191 ms 142.250.209.70 (142.250.209.70) 50.630 ms 192.178.110.104 (192.178.110.104) 55.105
12 216.239.48.65 (216.239.48.65) 40.470 ms 142.250.46.170 (142.250.46.170) 50.171 ms 172.253.51.137 (172.253.51.137) 68.494 ms
13 216.239.50.23 (216.239.50.23) 82.026 ms 172.253.66.106 (172.253.66.106) 70.685 ms 216.239.62.181 (216.239.62.181) 64.327 ms
14 192.178.83.221 (192.178.83.221) 78.419 ms 172.253.51.137 (172.253.51.137) 72.431 ms 142.251.52.229 (142.251.52.229) 77.782 ms
15 142.251.52.231 (142.251.52.231) 70.406 ms del12s02-in-f4.1e100.net (142.250.194.36) 79.307 ms 142.251.52.229 (142.251.52.229)
user@swlab-PC:~$
```

Number of Hosts = 15

When running a traceroute to www.google.com, we may notice that some intermediate hops display * * * instead of responding with latency times. This behaviour is particularly common with Google and other large tech companies for the Privacy and Security Policies.

Google prioritizes user privacy and network security. Google's infrastructure is protected by advanced firewalls that may block traceroute requests. Many of their routers and devices are configured to block or ignore traceroute probes (ICMP or UDP packets). This ensures protection against attacks since Exposing details about their internal network infrastructure could make it vulnerable to mapping and targeted attacks. By not responding, Google minimizes the amount of information available about its internal routing.

PART – 2: PACKET ANALYSIS

2.1) Analysis of DNS Packets: Structure and its Traffic:

a) The Domain Name System (DNS) for <https://www.iitkgp.ac.in> uses **User Datagram Protocol (UDP)** for communication, instead of the Transmission Control Protocol (TCP). The primary reason for this can be that UDP is a lightweight protocol with no need for connection setup (like TCP's handshake process). Also DNS queries are small, simple, and require quick responses. UDP's simplicity ensures minimal delay.

b) Source of the DNS Query: 10.5.16.134 This is the device initiating the DNS query.

Destination of the DNS Query: 172.16.1.180 This is the DNS server the query is sent to.

The query is for resolving the domain name www.iitkgp.ac.in to its corresponding IP address.

c) Only **1 query request** was sent from your browser (host machine) to DNS server during the name-to-IP resolution.

d) The DNS server at 172.16.1.180 responds with the actual IPv4 address.

e) One DNS server, 172.16.1.180, is involved. The DNS server successfully responds to the query.

f) Query Record:

Name: www.iitkgp.ac.in

Type: A (IPv4 Address)

Class: IN (Internet)

Answer Record:

Name: www.iitkgp.ac.in

Type: A (IPv4 Address)

Class: IN (Internet)

TTL: 86400 seconds (1 day)

Data Length: 4 bytes (IPv4 address)

Resolved IP Address: 172.16.3.10

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help



dns && (frame contains "iitkgp")

No.	Time	Source	Destination	Protocol	Length	Info
3982	6.431306897	10.5.16.134	172.16.1.180	DNS	87	Standard query 0x4a0d A www.iitkgp.ac.in OPT
3992	6.435033239	172.16.1.180	10.5.16.134	DNS	103	Standard query response 0x4a0d A www.iitkgp.ac.in A 172.16.3...

▶ Frame 3982: 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on interface eno1, id 0
 ▶ Ethernet II, Src: HewlettP_69:f0:2c (84:a9:3e:69:f0:2c), Dst: 90:88:55:9c:c4:4a (90:88:55:9c:c4:4a)
 ▶ Internet Protocol Version 4, Src: 10.5.16.134, Dst: 172.16.1.180
 ▶ User Datagram Protocol, Src Port: 36094, Dst Port: 53

▶ Domain Name System (query)
 Transaction ID: 0x4a0d
 ▶ Flags: 0x0100 Standard query
 Questions: 1
 Answer RRs: 0
 Authority RRs: 0
 Additional RRs: 1
 ▶ Queries
 ▶ www.iitkgp.ac.in: type A, class IN
 Name: www.iitkgp.ac.in
 [Name Length: 16]
 [Label Count: 4]
 Type: A (Host Address) (1)
 Class: IN (0x0001)
 ▶ Additional records
[\[Response In: 3992\]](#)

```

0000  90 88 55 9c c4 4a 84 a9 3e 69 f0 2c 08 00 45 00  ..U..J..>i...E.
0010  00 49 ba b0 40 00 40 11 b7 a4 0a 05 10 86 ac 10  .Y....=..h....
0020  01 b4 8c fe 00 35 00 35 c8 95 4a 0d 01 00 00 01  ..5...E..J....
0030  00 00 00 00 00 01 03 77 77 77 06 69 69 74 6b 67  ....w ww.iitkg
0040  70 02 61 63 02 69 6e 00 00 01 00 01 00 00 29 02  p.ac.in.....)
0050  00 00 00 00 00 00 00 00  .....
```

wireshark_eno1_20250114180641_FKHFYa.pcapng

Packets: 29922 · Displayed: 2 (0.0%)

Profile: Default

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help



dns && (frame contains "iitkgp")

No.	Time	Source	Destination	Protocol	Length	Info
3982	6.431306897	10.5.16.134	172.16.1.180	DNS	87	Standard query 0x4a0d A www.iitkgp.ac.in OPT
3992	6.435033239	172.16.1.180	10.5.16.134	DNS	103	Standard query response 0x4a0d A www.iitkgp.ac.in A 172.16.3...

▶ Domain Name System (response)
 Transaction ID: 0x4a0d
 ▶ Flags: 0x8500 Standard query response, No error
 Questions: 1
 Answer RRs: 1
 Authority RRs: 0
 Additional RRs: 1
 ▶ Queries
 ▶ www.iitkgp.ac.in: type A, class IN
 Name: www.iitkgp.ac.in
 [Name Length: 16]
 [Label Count: 4]
 Type: A (Host Address) (1)
 Class: IN (0x0001)
 ▶ Answers
 ▶ www.iitkgp.ac.in: type A, class IN, addr 172.16.3.10
 Name: www.iitkgp.ac.in
 Type: A (Host Address) (1)
 Class: IN (0x0001)
 Time to live: 86400 (1 day)
 Data length: 4
 Address: 172.16.3.10
 ▶ Additional records
[\[Request In: 3982\]](#)
[\[Time: 0.003726342 seconds\]](#)

```

0000  84 a9 3e 69 f0 2c 90 88 55 9c c4 4a 08 00 45 00  ..>i...U..J..E.
0010  00 59 b4 dd 00 00 3d 11 00 68 ac 10 01 b4 0a 05  .Y....=..h....
0020  10 86 00 35 8c fe 00 45 88 98 4a 0d 85 80 00 01  ..5...E..J....
0030  00 01 00 00 00 01 03 77 77 77 06 69 69 74 6b 67  ....w ww.iitkg
0040  70 02 61 63 02 69 6e 00 00 01 00 01 c0 0c 00 01  p.ac.in.....)
0050  00 01 00 01 51 80 00 04 ac 10 03 0a 00 00 29 04  ....Q.....)
0060  d0 00 00 00 00 00 00 00  .....
```

wireshark_eno1_20250114180641_FKHFYa.pcapng

Packets: 37092 · Displayed: 2 (0.0%)

Profile: Default

2.2) Web Traffic (HTTP)

a) The captured HTTP traffic between the client (source) and the web server (destination):

- | | |
|--|-------------------------------|
| • GET /~grovesd/: Request for the main webpage. | Response: 200 OK (text/html). |
| • GET /~grovesd/css/base.css: Request for the CSS file. | Response: 200 OK (text/css). |
| • GET /~grovesd/images/big-bear.png: Request for an image. | Response: 200 OK (PNG). |
| • GET /favicon.ico: Request for the website's favicon. | Response: 404 Not Found. |

b) HTTP Headers Analysis:

Wireshark capture showing HTTP traffic between swlab-PC.local and web.simmons.edu. The packet list shows four requests:

No.	Time	Source	Destination	Protocol	Length	Info
1017	39.558220822	swlab-PC.local	web.simmons.edu	HTTP	498	GET /~grovesd/ HTTP/1.1
1021	39.794274597	web.simmons.edu	swlab-PC.local	HTTP	2321	HTTP/1.1 200 OK (text/html)
1027	39.826571082	swlab-PC.local	web.simmons.edu	HTTP	406	GET /~grovesd/css/base.css HTTP/1.1
1035	40.059200504	web.simmons.edu	swlab-PC.local	HTTP	2510	HTTP/1.1 200 OK (text/css)
1039	40.072204729	swlab-PC.local	web.simmons.edu	HTTP	471	GET /~grovesd/images/big-bear.png HTTP/1.1
1091	40.538455947	web.simmons.edu	swlab-PC.local	HTTP	808	HTTP/1.1 200 OK (PNG)
1182	40.842690309	swlab-PC.local	web.simmons.edu	HTTP	442	GET /favicon.ico HTTP/1.1
1189	41.072678957	web.simmons.edu	swlab-PC.local	HTTP	462	HTTP/1.1 404 Not Found (text/html)

Frame 1021: 498 bytes on wire (3984 bits), 498 bytes captured (3984 bits) on interface eno1, id 0
 Ethernet II, Src: swlab-PC.local (84:a9:3e:69:f0:2c), Dst: gateway (90:88:55:9c:c4:4a)
 Internet Protocol Version 4, Src: swlab-PC.local (10.5.16.134), Dst: web.simmons.edu (69.43.111.82)
 Transmission Control Protocol, Src Port: 37248, Dst Port: 80, Seq: 1, Ack: 1, Len: 432
 Hypertext Transfer Protocol
 GET /~grovesd/ HTTP/1.1\r\n
 [Expert Info (Chat/Sequence): GET /~grovesd/ HTTP/1.1\r\n]
 Request Method: GET
 Request URI: /~grovesd/
 Request Version: HTTP/1.1
 Host: web.simmons.edu\r\n
 Connection: keep-alive\r\n
 Upgrade-Insecure-Requests: 1\r\n
 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/97.0.4692.71 Safari/537.36\r\n
 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9\r\n
 Accept-Encoding: gzip, deflate\r\n
 Accept-Language: en-GB,en;q=0.9\r\n
 \r\n
 [Full request URI: http://web.simmons.edu/~grovesd/]
 [HTTP request 1/4]
 [Response in frame: 1021]
 [Next request in frame: 1027]

Hex dump of the captured packet (frame 1021):

```

0000  90 88 55 9c c4 4a 84 a9 3e 69 f0 2c 08 00 45 00  --U..J..1,--E-
0010  01 e4 ab ed 40 00 40 06 be 1e 0a 05 10 86 45 2b  --@@-.....E+
0020  6f 52 91 80 00 50 8a d2 30 82 d9 f7 c5 58 80 18  oR...P..0...X..
0030  01 f6 d0 de 00 00 01 01 08 0a 48 47 5b 3c f5 a1  --.....HG[<..
0040  f3 b0 47 45 54 20 2f 7e 67 72 6f 76 65 73 64 2f  --GET /~ grovesd/
0050  20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a  HTTP/1.1 Host:
0060  20 77 65 62 2e 73 69 6d 6d 6f 6e 73 2e 65 64 75  web.sim mons.edu
0070  0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65  --Connec tion: ke
0080  65 70 2d 61 6c 69 76 65 0d 0a 55 70 67 72 61 64  ep-alive --Upgrad
0090  65 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65  e-Insecu re-Reque
00a0  73 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65  sts: 1-- User-Age
  
```

HTTP Request Header for GET /~grovesd/:

- Method: GET
- Request URI: /~grovesd/
- Version: HTTP/1.1
- Host: web.simmons.edu
- User-Agent: Describes the browser (e.g., Chrome/97.0).
- Connection: keep-alive

HTTP Response Header for GET /~grovesd/:

- Status Code: 200 OK
- Content-Type: text/html

c) To fully load the webpage, 8 HTTP packets (4 requests + 4 responses) were exchanged.

Total Requests: 4

- HTML document (1 request)
- CSS file (1 request)
- Image (1 request)
- Favicon (1 request)

Total Responses: 4

- HTML document (200 OK)
- CSS file (200 OK)
- Image file (200 OK)
- Favicon (404 Not Found)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

(ip.addr eq 10.5.16.134 and ip.addr eq 69.43.111.82) and http

No.	Time	Source	Destination	Protocol	Length	Info
1017	39.558220022	swlab-PC.local	web.simmons.edu	HTTP	498	GET /-grovesd/ HTTP/1.1
1021	39.794274597	web.simmons.edu	swlab-PC.local	HTTP	2321	HTTP/1.1 200 OK (text/html)
1027	39.826571082	swlab-PC.local	web.simmons.edu	HTTP	406	GET /-grovesd/css/base.css HTTP/1.1
1035	40.059200504	web.simmons.edu	swlab-PC.local	HTTP	2510	HTTP/1.1 200 OK (text/css)
1039	40.072204729	swlab-PC.local	web.simmons.edu	HTTP	471	GET /-grovesd/images/big-bear.png HTTP/1.1
1091	40.538455947	web.simmons.edu	swlab-PC.local	HTTP	808	HTTP/1.1 200 OK (PNG)
1182	40.842690309	swlab-PC.local	web.simmons.edu	HTTP	442	GET /favicon.ico HTTP/1.1
1189	41.072678957	web.simmons.edu	swlab-PC.local	HTTP	462	HTTP/1.1 404 Not Found (text/html)

Frame 1021: 2321 bytes on wire (18568 bits), 2321 bytes captured (18568 bits) on interface eno1, id 0
 Ethernet II, Src: gateway (90:88:55:9c:c4:4a), Dst: swlab-PC.local (84:a9:3e:69:f0:2c)
 Internet Protocol Version 4, Src: web.simmons.edu (69.43.111.82), Dst: swlab-PC.local (10.5.16.134)
 Transmission Control Protocol, Src Port: 80, Dst Port: 37248, Seq: 1, Ack: 433, Len: 2255

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n
 Date: Tue, 14 Jan 2025 12:39:04 GMT\r\n
 Server: Apache\r\n
 Last-Modified: Tue, 03 Sep 2019 00:33:59 GMT\r\n
 ETag: "7b2-5919b3e8debc0"\r\n
 Accept-Ranges: bytes\r\n
 Content-Length: 1970\r\n
 Keep-Alive: timeout=5, max=100\r\n
 Connection: Keep-Alive\r\n
 Content-Type: text/html; charset=UTF-8\r\n
 \r\n
 [HTTP response 1/4]
 [Time since request: 0.236054575 seconds]
 [Request in frame: 1017]
 [Next request in frame: 1027]
 [Next response in frame: 1035]
 [Request URI: http://web.simmons.edu/favicon.ico]
 File Data: 1970 bytes

Line-based text data: text/html (57 lines)

```

0000 84 a9 3e 69 f0 2c 00 88 55 9c c4 4a 08 00 45 20 -->1...U...E
0010 09 03 2f 6a 40 00 2a 06 49 63 45 2b 6f 52 0a 05 --/j...Ic+oR...
0020 10 86 00 50 91 80 d9 f7 c5 58 8a d2 32 32 80 18 --P...X-22...
0030 00 f9 d7 fd 00 00 01 01 08 0a f5 a1 f4 99 48 47 --.....HG
0040 5b 3c 48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f [<HTTP/1.1 200 0
0050 4b 0d 0a 44 61 74 65 3a 20 54 75 65 2c 20 31 34 K-Date: Tue, 14
0060 20 4a 61 6e 20 32 30 32 35 20 31 32 3a 33 39 3a Jan 202 5 12:39:
0070 30 34 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 04 GMT- Server:
0080 41 70 61 63 68 65 0d 0a 4c 61 73 74 2d 4d 6f 64 Apache- Last-Mod
0090 69 66 69 65 64 3a 20 54 75 65 2c 20 30 33 20 53 ified: T ue, 03 S
00a0 65 70 20 32 30 31 39 20 30 30 3a 33 33 3a 35 39 ep 2019 00:33:59
  
```

Hypertext Transfer Protocol: Protocol Packets: 3788 - Displayed: 8 (0.2%) Profile: Default

2.3) ICMP Traffic (Ping/Traceroute)

a) My PC IP Address: 10.5.16.134

Friend's PC IP Address: 10.5.16.135

PING: Requests and replies are paired, confirming a successful ICMP communication. Sequence numbers increase by increments (e.g., 1, 2, 3, etc.). TTL (Time To Live) is set to 64, standard for ping responses in local networks.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4860	263.213514583	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=1/256, ttl=64 (reply in 4861)
4861	263.213555715	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 4860)
4868	264.251569091	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=2/512, ttl=64 (reply in 4869)
4869	264.251607454	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 4868)
4873	265.274672997	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=3/768, ttl=64 (reply in 4874)
4874	265.274712284	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 4873)
4889	266.298387522	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=4/1024, ttl=64 (reply in 4890)
4890	266.298425139	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 4889)
4898	267.322491913	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=5/1280, ttl=64 (reply in 4899)
4899	267.322530587	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=5/1280, ttl=64 (request in 4898)
4913	268.346769157	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=6/1536, ttl=64 (reply in 4914)
4914	268.346806722	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=6/1536, ttl=64 (request in 4913)
4947	269.370661445	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=7/1792, ttl=64 (reply in 4948)
4948	269.370698828	10.5.16.134	10.5.16.135	ICMP	98	Echo (ping) reply id=0x0001, seq=7/1792, ttl=64 (request in 4947)
4954	270.394627744	10.5.16.135	10.5.16.134	ICMP	98	Echo (ping) request id=0x0001, seq=8/2048, ttl=64 (reply in 4955)

Frame 4860: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eno1, id 0
 Ethernet II, Src: HewlettP-69:f0:8f (84:a9:3e:69:f0:8f), Dst: swlab-PC.local (84:a9:3e:69:f0:2c)
 Internet Protocol Version 4, Src: 10.5.16.135, Dst: 10.5.16.134

Internet Control Message Protocol

Type: 8 (Echo (ping) request)
 Code: 0
 Checksum: 0xf9d3 [correct]
 [Checksum Status: Good]
 Identifier (BE): 1 (0x0001)
 Identifier (LE): 256 (0x0100)
 Sequence number (BE): 1 (0x0001)
 Sequence number (LE): 256 (0x0100)
 [Response frame: 4861]
 Timestamp from icmp data: Jan 14, 2025 18:12:47.000000000 IST
 [Timestamp from icmp data (relative): 0.708369173 seconds]
 Data (48 bytes)
 Data: e7930a0000000000101112131415161718191a1b1c1d1e1f...
 [Length: 48]

```

0000 84 a9 3e 69 f0 2c 84 a9 3e 69 f0 8f 08 00 45 00 -->1...>1...E-
0010 00 54 96 ba 40 00 40 01 6e d8 0a 05 10 87 0a 05 --T...@...n...
0020 10 86 00 08 f9 d3 00 01 00 01 c7 5b 80 67 00 00 --.....[:g...
0030 00 00 e7 93 0a 00 00 00 00 00 10 11 12 13 14 15 --.....I"#S%
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 --.....&'()*+,-./012345
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 --.....
0060 36 37 38 39 3a 3b 3c 3d 3e 3f 40 41 42 43 44 45 --.....
  
```

Internet Control Message Protocol (icmp), 64 bytes Packets: 6272 - Displayed: 52 (0.8%) Profile: Default

The image shows a Wireshark packet capture of ICMP messages. The top pane displays a list of packets, all of which are ICMP Destination Unreachable messages from 10.5.16.135 to 10.5.16.134. The middle pane shows the details of the selected packet (Frame 79), including the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol (ICMP) details. The ICMP details show a Type of 3 (Destination unreachable) and a Code of 3 (Port unreachable). The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
79	2.883695907	10.5.16.135	10.5.16.134	ICMP	102	Destination unreachable (Port unreachable)
80	2.883696116	10.5.16.135	10.5.16.134	ICMP	102	Destination unreachable (Port unreachable)
81	2.883757029	10.5.16.135	10.5.16.134	ICMP	102	Destination unreachable (Port unreachable)
82	2.883757157	10.5.16.135	10.5.16.134	ICMP	102	Destination unreachable (Port unreachable)
83	2.883757264	10.5.16.135	10.5.16.134	ICMP	102	Destination unreachable (Port unreachable)
84	2.883757384	10.5.16.135	10.5.16.134	ICMP	102	Destination unreachable (Port unreachable)

Frame 79: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface eno1, id 0
 Ethernet II, Src: HewlettP_69:f0:8f (84:a9:3e:69:f0:8f), Dst: HewlettP_69:f0:2c (84:a9:3e:69:f0:2c)
 Internet Protocol Version 4, Src: 10.5.16.135, Dst: 10.5.16.134
 Internet Control Message Protocol
 Type: 3 (Destination unreachable)
 Code: 3 (Port unreachable)
 Checksum: 0x324d [correct]
 [Checksum Status: Good]
 Unused: 00000000
 Internet Protocol Version 4, Src: 10.5.16.134, Dst: 10.5.16.135
 User Datagram Protocol, Src Port: 53560, Dst Port: 33434
 Source Port: 53560
 Destination Port: 33434
 Length: 40
 Checksum: 0x81af [unverified]
 [Checksum Status: Unverified]
 [Stream index: 3]
 Data (32 bytes)
 Data: 404142434445464748494a4b4c4d4e4f50515253545556575859...
 [Length: 32]

eno1: <live capture in progress> Packets: 211 · Displayed: 6 (2.8%) Profile: Default

TRACEROUTE: Traceroute sends packets with a Time-to-Live (TTL) value. The TTL starts small (e.g., 1) and increases with each packet. Every router the packet passes through reduces the TTL by 1. When the packet reaches the destination, it sends back an ICMP Port Unreachable message if UDP packets are used. This tells Traceroute that the destination has been reached.

```
user@swlab-PC:~$ ping 10.5.16.135
PING 10.5.16.135 (10.5.16.135) 56(84) bytes of data:
64 bytes from 10.5.16.135: icmp_seq=1 ttl=64 time=0.489 ms
64 bytes from 10.5.16.135: icmp_seq=2 ttl=64 time=0.659 ms
64 bytes from 10.5.16.135: icmp_seq=3 ttl=64 time=0.649 ms
64 bytes from 10.5.16.135: icmp_seq=4 ttl=64 time=0.666 ms
64 bytes from 10.5.16.135: icmp_seq=5 ttl=64 time=0.641 ms
64 bytes from 10.5.16.135: icmp_seq=6 ttl=64 time=0.664 ms
64 bytes from 10.5.16.135: icmp_seq=7 ttl=64 time=0.391 ms
64 bytes from 10.5.16.135: icmp_seq=8 ttl=64 time=0.646 ms
64 bytes from 10.5.16.135: icmp_seq=9 ttl=64 time=0.642 ms
64 bytes from 10.5.16.135: icmp_seq=10 ttl=64 time=0.663 ms
64 bytes from 10.5.16.135: icmp_seq=11 ttl=64 time=0.649 ms
64 bytes from 10.5.16.135: icmp_seq=12 ttl=64 time=0.410 ms
64 bytes from 10.5.16.135: icmp_seq=13 ttl=64 time=0.650 ms
64 bytes from 10.5.16.135: icmp_seq=14 ttl=64 time=0.655 ms
^C
--- 10.5.16.135 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13316ms
rtt min/avg/max/mdev = 0.391/0.605/0.666/0.093 ms
user@swlab-PC:~$ traceroute 10.5.16.135
traceroute to 10.5.16.135 (10.5.16.135), 30 hops max, 60 byte packets
 1  10.5.16.135 (10.5.16.135)  0.699 ms  0.657 ms  0.625 ms
```

b) PING: ICMP Echo Request packets are sent to the unreachable IP address (e.g., 192.168.31.3 in this case). These packets attempt to reach the destination host. Since the host does not exist, there is no reply (no ICMP Echo Reply packets). ICMP Echo Request packets are continuously sent with increasing sequence numbers (seq=15, seq=16, etc.). The field **[No response seen]** is present, indicating no ICMP Echo Reply packets were received. The network or host is unreachable, so no ICMP Echo Reply or Destination Unreachable response is received.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
1839	85.150113122	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=14/3584, ttl=64 (no response found..)
1841	86.174111510	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=15/3840, ttl=64 (no response found..)
1864	87.198000697	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=16/4096, ttl=64 (no response found..)
1867	88.222052463	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=17/4352, ttl=64 (no response found..)
1903	89.245083389	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=18/4608, ttl=64 (no response found..)
1905	90.269007351	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=19/4864, ttl=64 (no response found..)
1931	91.294033138	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=20/5120, ttl=64 (no response found..)
1945	92.318115028	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=21/5376, ttl=64 (no response found..)
1959	93.342117712	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=22/5632, ttl=64 (no response found..)
1995	94.366003200	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=23/5888, ttl=64 (no response found..)
2003	95.390004866	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=24/6144, ttl=64 (no response found..)
2011	96.413998493	10.5.16.134	192.168.31.3	ICMP	98	Echo (ping) request id=0x0002, seq=25/6400, ttl=64 (no response found..)

Frame 1841: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eno1, id 0
 Ethernet II, Src: HewlettP_69:f0:2c (84:a9:3e:69:f0:2c), Dst: 90:88:55:9c:c4:4a (90:88:55:9c:c4:4a)
 Internet Protocol Version 4, Src: 10.5.16.134, Dst: 192.168.31.3
 Internet Control Message Protocol
 Type: 8 (Echo (ping) request)
 Code: 0
 Checksum: 0x53d1 [correct]
 [Checksum Status: Good]
 Identifier (BE): 2 (0x0002)
 Identifier (LE): 512 (0x0200)
 Sequence number (BE): 15 (0x000f)
 Sequence number (LE): 3840 (0x0f00)
 [No response seen]
 Timestamp from icmp data: Jan 16, 2025 11:26:12.000000000 IST
 [Timestamp from icmp data (relative): 0.803830476 seconds]
 Data (48 bytes)
 Data: d4430c000000000010112131415161718191a1b1c1d1e1f...
 [Length: 48]

eno1: <live capture in progress> Packets: 2032 - Displayed: 34 (1.7%) Profile: Default

TRACEROUTE: The traceroute stops at the last reachable router, which sends an ICMP Destination Unreachable message. Traceroute still sends packets with increasing TTL values. Routers along the path respond with **ICMP Time Exceeded** messages. Eventually, a router detects that the destination is unreachable and sends an **ICMP Destination**

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
297	33.823228269	10.120.2.33	10.5.16.134	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
298	33.823228375	10.5.16.2	10.5.16.134	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
299	33.823228409	10.120.2.33	10.5.16.134	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
300	33.823228443	10.5.16.2	10.5.16.134	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
301	33.823228479	10.120.2.33	10.5.16.134	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
302	33.823228513	10.5.16.2	10.5.16.134	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
304	33.825338931	10.255.1.3	10.5.16.134	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
305	33.826665031	10.255.1.3	10.5.16.134	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
306	33.826807146	10.255.1.3	10.5.16.134	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)

Frame 297: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface eno1, id 0
 Ethernet II, Src: 90:88:55:9c:c4:4a (90:88:55:9c:c4:4a), Dst: HewlettP_69:f0:2c (84:a9:3e:69:f0:2c)
 Internet Protocol Version 4, Src: 10.120.2.33, Dst: 10.5.16.134
 Internet Control Message Protocol
 Type: 11 (Time-to-live exceeded)
 Code: 0 (Time to live exceeded in transit)
 Checksum: 0xe474 [correct]
 [Checksum Status: Good]
 Unused: 00000000
 Internet Protocol Version 4, Src: 10.5.16.134, Dst: 192.168.31.3
 User Datagram Protocol, Src Port: 48690, Dst Port: 33437
 Source Port: 48690
 Destination Port: 33437
 Length: 40
 Checksum: 0xc992 [unverified]
 [Checksum Status: Unverified]
 [Stream index: 20]

eno1: <live capture in progress> Packets: 506 - Displayed: 9 (1.8%) Profile: Default

c) Traceroute operations differ significantly for reachable and unreachable hosts due to their responses at each hop. For reachable hosts, traceroute completes successfully, as each intermediate router and the destination host respond with ICMP messages, providing latency and routing path information.

However, for unreachable hosts, intermediate routers forward the packets, but once they reach a network segment without the specified host, the destination remains unresponsive. This results in traceroute displaying stars (* * *) for subsequent hops, indicating no response. In both cases, traceroute relies on ICMP packets with incremented TTL values to trace the route, but the absence of responses in the unreachable host case highlights the lack of a valid endpoint in the specified network. Even after 30 hops, the TTL packet couldn't find a destination, which indicates the host is unreachable in that network.

Parameter	Reachable Host	Unreachable Host
Target Host IP	10.5.16.135	192.168.31.3
Response Type	ICMP Port Unreachable (Type 3, Code 3)	No Response Seen
Hop Count	1	30 (since it reached max limit)
Final Destination Reached	Yes	No
ICMP Type at Final Hop	Type 3 (Port Unreachable)	Type 11 (TTL Exceeded)

Reachable Host:

```
user@swlab-PC:~$ traceroute 10.5.16.135
traceroute to 10.5.16.135 (10.5.16.135), 30 hops max, 60 byte packets
 1  10.5.16.135 (10.5.16.135)  0.699 ms  0.657 ms  0.625 ms
user@swlab-PC:~$
```

Unreachable Host:

```
user@swlab-PC:~$ traceroute 192.168.31.3
traceroute to 192.168.31.3 (192.168.31.3), 30 hops max, 60 byte packets
 1  _gateway (10.5.16.2)  0.417 ms  0.391 ms  0.377 ms
 2  10.120.2.33 (10.120.2.33)  0.364 ms  0.349 ms  0.336 ms
 3  10.255.1.3 (10.255.1.3)  3.758 ms  2.419 ms  3.875 ms
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```